

Case study: differential gene expression analysis

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This vignette illustrates the use of `quarto` to build a reproducible data analysis report.

```
require("ggplot2") || install.packages("ggplot2")
require("sanssouci") || remotes::install_github("sanssouci-org/sanssouci@develop")
```

```
library("ggplot2")
library("sanssouci")
library("matrixStats")
```

1 Motivation: a differential gene expression study

Differential gene expression studies aim at identifying genes whose mean expression level differs significantly between two (or more) populations, based on a sample of gene expression measurements from individuals from these populations. The data is stored in the package `sanssouci.data` and can be loaded as follows:

```
data("RNAseq_blca", package = "sanssouci.data")
Y <- RNAseq_blca
groups <- ifelse(colnames(RNAseq_blca) == "III", 1, 0)
rm(RNAseq_blca)
```

This data set consists of gene expression measurements for $n = 270$ patients from the Cancer Genome Atlas Urothelial Bladder Carcinoma (TCGA-BLCA) data collection (Cancer Genome Atlas Research Network et al. (2014)). The patients are classified into two subgroups, according to their clinical stage (II or III).

2 Preprocessing

Following standard practice in RNAseq data analysis, we convert these measurements to *count per millions* (CPM) in order to make them comparable across samples.

```
CPM <- Y/colSums(Y)*1e6  
plot(density(log(1 + CPM)))
```

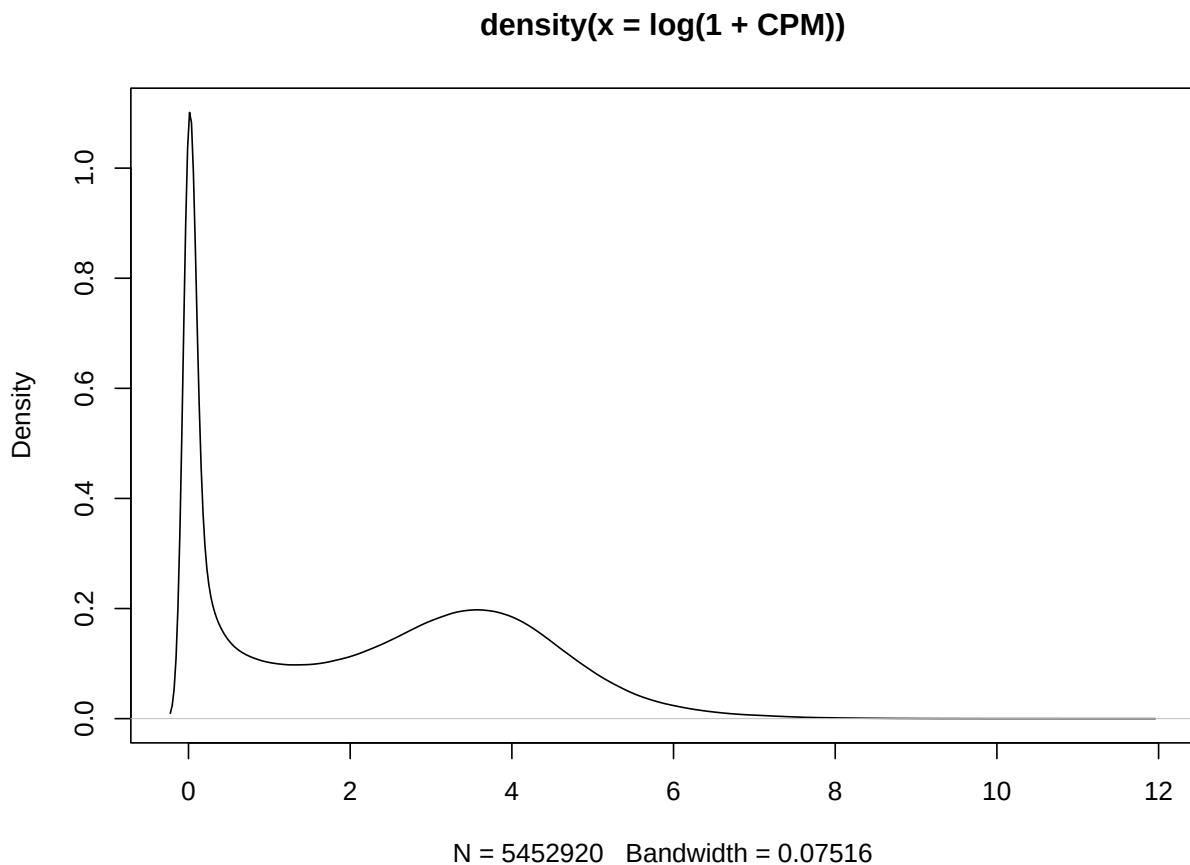


Figure 1: Empirical distribution of gene expression across all genes and samples

We now filter out genes with very low counts in all samples and plot the corresponding distribution:

```
ww <- which(rowQuantiles(log(1 + CPM), prob = 0.75) < log(1 + 5))  
Y <- CPM[-ww, ]  
plot(density(log(1 + Y)))  
m <- nrow(Y)
```

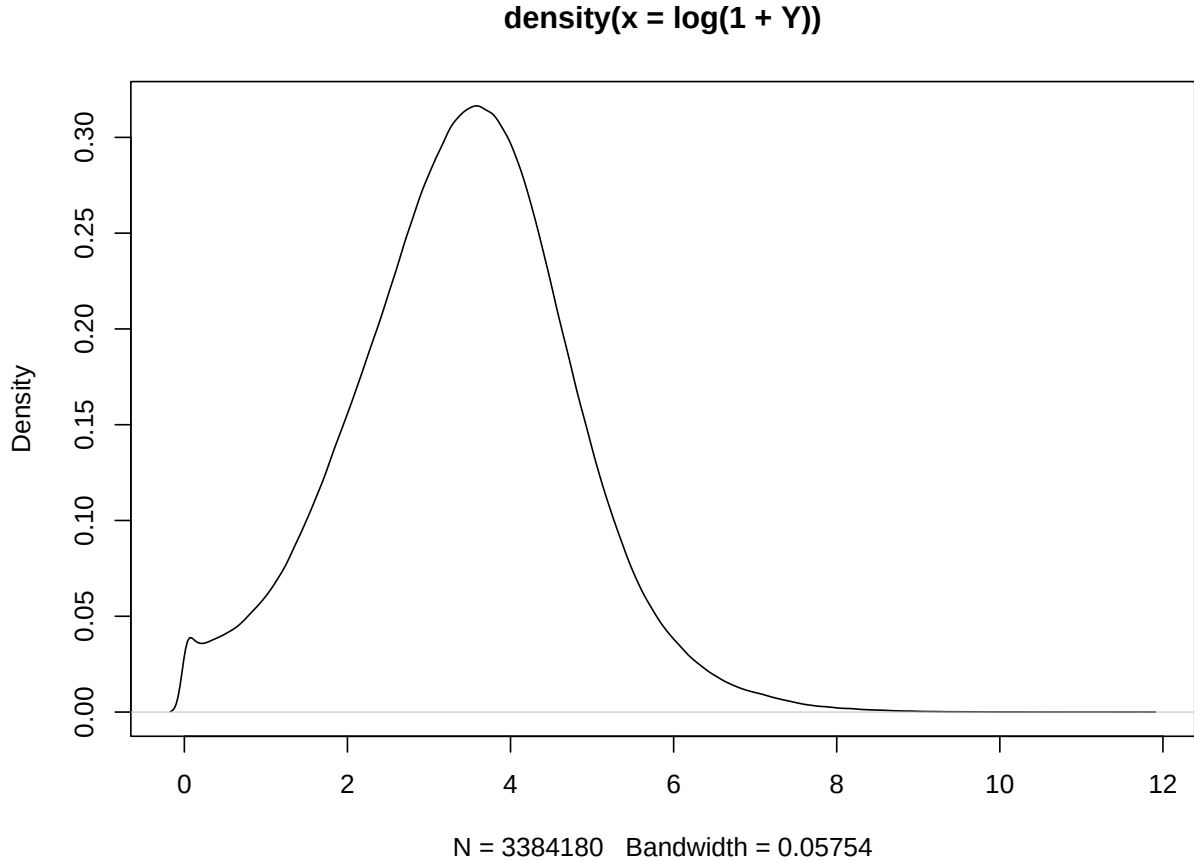


Figure 2: Empirical distribution of gene expression across all samples after gene filtering

The first mode in Figure 1 has been filtered out in Figure 2.

3 Differential analysis

The goal of this study is to understand the molecular differences at the gene expression level between the stage II and stage III populations. Each patient is characterized by a vector of $m = 12534$ gene expression values. We aim at addressing the following question:

For which genes is there a difference in the distribution of expression levels between the stage II and stage III populations?

This question can be addressed by performing one statistical test per gene, and to define “differentially expressed” genes as those passing some significance threshold. Below, we use the Wilcoxon rank sum (or Mann-Whitney) test for comparing two independent samples.

This is a rank-based non-parametric test. A fast version of this test is implemented in the `sanssouci::rowWilcoxonTests` function.

We create an object of class `SansSouci` and run the `fit` method to use the (parametric) Simes method introduced by Goeman and Solari (2011):

```
obj <- SansSouci(Y = log(1 + Y), groups = groups)
alpha = 0.1
res_Simes <- fit(obj, B = 0, family = "Simes", alpha = alpha,
                rowTestFUN = rowWilcoxonTests) ## B=0 => no calibration!
```

How many genes are differentially expressed? For any user-defined gene subset, the method reports a lower bound on the number of truly differentially expressed genes (DEG), or, equivalently, an upper bound on the proportion of truly differentially expressed genes.

The target risk level is set to $\alpha = 0.1$, meaning that all of the statements made below will be true with probability greater than $1 - \alpha = 0.9$.

For example, we can consider the following three gene subsets:

- S_1 : all genes
- S_2 : the 500 most significant genes
- S_3 : all genes whose p -value is below 0.001 and fold change above 0.1:

```
S1 <- 1:m
p1 <- predict(res_Simes, S = S1)

S2 <- which(rank(pValues(res_Simes)) <= 500)
p2 <- predict(res_Simes, S = S2)

S3 <- which(pValues(res_Simes) < 0.001 & abs(foldChanges(res_Simes)) > 0.1)
p3 <- predict(res_Simes, S = S3)
```

Table 1 summarizes the results in a single table:

```
pred <- rbind(p1, p2, p3)
labels <- c("all genes",
           "500 most significant genes",
           "all genes whose $p$-value is below 0.001 and fold change above 0.1")
pred <- data.frame("Gene subset" = labels, pred)
knitr::kable(pred)
```

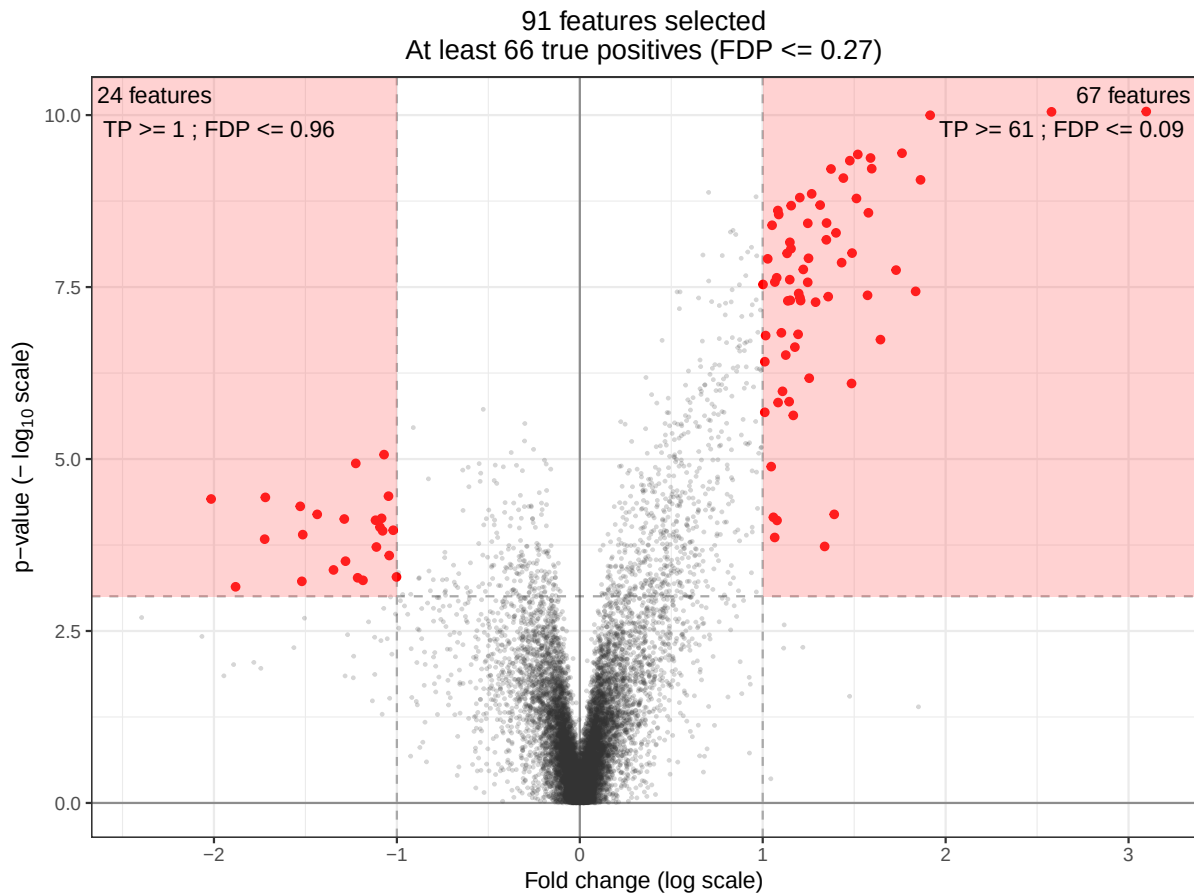
Table 1: Confidence bounds on the number of truly differentially expressed genes (DEG) for three gene subsets: number of true positives (TP) and proportion of false positives (FDP, False Discovery Proportion)

	Gene.subset	TP	FDP
p1	all genes	928	0.9259614
p2	500 most significant genes	479	0.0420000
p3	all genes whose p -value is below 0.001 and fold change above 0.1	766	0.1393258

4 Volcano plots

A classical practice in gene expression studies is to define DE genes as those passing both a significance filter (small p -value) and an effect size or “fold change” filter. Here, the fold change of a gene is defined as the difference between the expression medians (on the log scale) of the two groups compared. This double selection by p -value and fold change corresponds to two sets of genes, with positive/negative fold change, which can be represented in the following plot:

```
volcanoPlot(res_Simes, p = 1e-3, r = 1)
```



This type of plot is called a “volcano plot” Cui and Churchill (2003). Post hoc inference makes it possible to obtain statistical guarantees on selections such as the ones represented in the above figure.

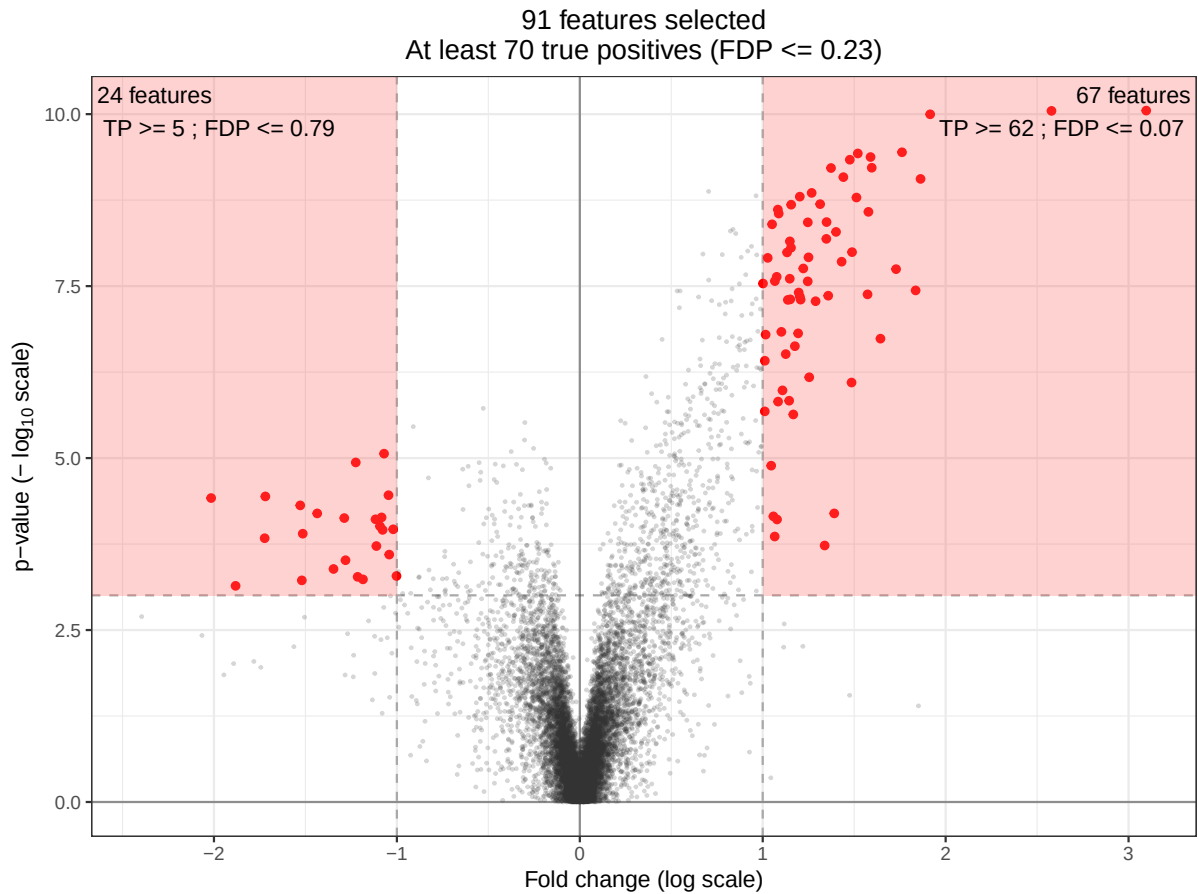
5 MA more powerful method

To obtain more informative results, we use the calibration method introduced by Blanchard et al. (2020) and Enjalbert Courrech and Neuvial (2022). This method uses sample label permutations to learn the gene dependence structure and obtain a more powerful method.

```
obj <- SansSouci(Y = log(1 + Y), groups = groups)
res_perm <- fit(obj, B = 1000, alpha = alpha, family = "Simes",
               rowTestFUN = rowWilcoxonTests)
```

The annotations in the corresponding volcano plot illustrate a gain in power (more true positives, lower FDP):

```
volcanoPlot(res_perm, p = 1e-3, r = 1)
```



6 Session information

```
sessionInfo()
#> R version 4.6.1 (2026-06-24)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p0.3.26.so; LAPACK version 3.11.0
#>
#> locale:
```

```

#> [1] LC_CTYPE=fr_FR.UTF-8      LC_NUMERIC=C
#> [3] LC_TIME=fr_FR.UTF-8        LC_COLLATE=fr_FR.UTF-8
#> [5] LC_MONETARY=fr_FR.UTF-8    LC_MESSAGES=fr_FR.UTF-8
#> [7] LC_PAPER=fr_FR.UTF-8      LC_NAME=C
#> [9] LC_ADDRESS=C              LC_TELEPHONE=C
#> [11] LC_MEASUREMENT=fr_FR.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: Europe/Paris
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats      graphics  grDevices  datasets  utils      methods    base
#>
#> other attached packages:
#> [1] matrixStats_1.5.0  sanssouci_0.16.3  ggplot2_4.0.3
#>
#> loaded via a namespace (and not attached):
#> [1] Matrix_1.7-5      gtable_0.3.6      jsonlite_2.0.0
#> [4] dplyr_1.2.1        compiler_4.6.1     renv_1.1.4
#> [7] tidyselect_1.2.1   tinytex_0.60       scales_1.4.0
#> [10] yaml_2.3.12        fastmap_1.2.0      lattice_0.22-9
#> [13] R6_2.6.1           labeling_0.4.3     generics_0.1.4
#> [16] knitr_1.51         tibble_3.3.1       pillar_1.11.1
#> [19] RColorBrewer_1.1-3 rlang_1.3.0        xfun_0.60
#> [22] S7_0.2.2           otel_0.2.0         cli_3.6.6
#> [25] withr_3.0.3        magrittr_2.0.5     matrixTests_0.2.3.1
#> [28] digest_0.6.39      grid_4.6.1         rstudioapi_0.19.0
#> [31] lifecycle_1.0.5    vctrs_0.7.3        evaluate_1.0.5
#> [34] glue_1.8.1         farver_2.1.2       codetools_0.2-20
#> [37] rmarkdown_2.31     tools_4.6.1        pkgconfig_2.0.3
#> [40] htmltools_0.5.9

```

7 Reproducibility

To re-build this vignette from its source, use:

```

quarto::quarto_render("post-hoc_differential-expression_RNAseq.qmd", output_format = "pdf")
# To keep intermediate files, add option 'clean = FALSE' (rmarkdown-based rendering)
quarto::quarto_render("post-hoc_differential-expression_RNAseq.qmd", output_format = "html")

```

References

- Blanchard, Gilles, Pierre Neuvial, and Etienne Roquain. 2020. “Post Hoc Confidence Bounds on False Positives Using Reference Families.” *Annals of Statistics* 48 (3): 1281–303. <https://projecteuclid.org/euclid.aos/1594972818>.
- Cancer Genome Atlas Research Network et al. 2014. “Comprehensive Molecular Characterization of Urothelial Bladder Carcinoma.” *Nature* 507 (7492): 315.
- Cui, Xiangqin, and Gary A Churchill. 2003. “Statistical Tests for Differential Expression in cDNA Microarray Experiments.” *Genome Biol* 4 (4): 210.
- Enjalbert Courrech, Nicolas, and Pierre Neuvial. 2022. “Powerful and interpretable control of false discoveries in differential expression studies.” *Bioinformatics* 38 (23): 5214–21. <https://doi.org/10.1093/bioinformatics/btac693>.
- Goeman, Jelle J, and Aldo Solari. 2011. “Multiple Testing for Exploratory Research.” *Statistical Science* 26 (4): 584–97.